User Behavior & Funnel Analysis for a SaaS Product using SQL

11 SQL Queries for Business Insights

SUSMITTHA B

ANALYSIS OVERVIEW

1. **User Registration Sources** Analyze where users are signing up from.
2. **Signup Trends Per Day** Track daily user registration patterns.
3. **Common User Actions** Identify frequently performed actions by users.
4. **Active Users (Login Count)** Measure user engagement based on login frequency.
5. **User Dropoff Analysis** Pinpoint stages where users disengage from the product.
6. **Funnel Analysis - Stage Breakdown** Break down the user journey through key stages.
7. **Conversion Rate (Signup to Purchase)** Calculate the percentage of users converting from signup to purchase.
8. **Purchases by Source** Determine which registration sources lead to purchases.
9. **Days to Purchase** Measure the average time it takes for users to make a purchase after signing up.
10. **Signup to Purchase Rate** Evaluate the overall effectiveness of the signup process in driving sales.
11. **User Retention (7-day)** Analyze how many users return to the product after 7 days.

Query 1: User Registration Sources

```
SELECT source, COUNT(*) AS users_count
FROM users
GROUP BY source;
```

Result Grid | Filter Rows:

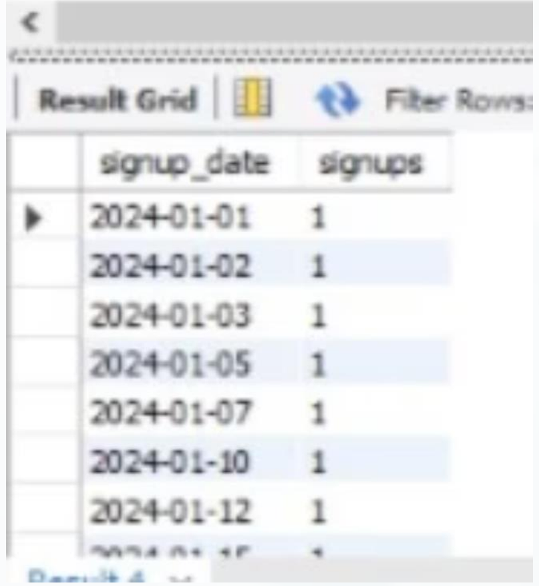
source	users_count
ads	19
organic	21
referral	10

Insight

Organic leads with 41% of signups (21 users), followed by paid ads at 35% (19 users). Referrals lag at 24% (10 users).

Query 2: Signup Trends Per Day

```
SELECT signup_date, COUNT(*) AS signups
FROM users
GROUP BY signup_date
ORDER BY signup_date;
```



signup_date	signups
2024-01-01	1
2024-01-02	1
2024-01-03	1
2024-01-05	1
2024-01-07	1
2024-01-10	1
2024-01-12	1

Insight

Consistent daily signups of 1 user per day show steady, predictable growth.

Query 3: Common User Actions

```
SELECT event_type, COUNT(*) AS event_count
FROM events
GROUP BY event_type
ORDER BY event_count DESC;
```

Result Grid | Filter Rows:

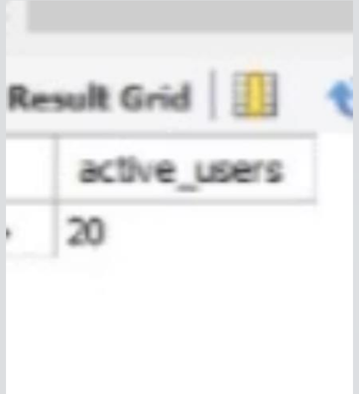
event_type	event_count
signup	35
login	29
purchase	10
add_to_cart	9

Insight

Signup events lead at 35, followed by logins at 29 (83% activation). Only 9 add-to-cart and 10 purchases indicate low purchase intent and conversion is the main challenge.

Query 4: Active Users (Login Count)

```
SELECT COUNT(DISTINCT user_id) AS active_users
FROM events
WHERE event_type = 'login';
```



Insight

20 out of ~50 users are active (40% activation rate). This indicates moderate engagement and suggests room for improvement in user onboarding .

Query 5: User Dropoff Analysis

```
SELECT u.user_id
FROM users u
LEFT JOIN events e
ON u.user_id = e.user_id AND e.event_type = 'login'
WHERE e.user_id IS NULL;
```

Result Grid

user_id
106
113
117
121
125
126
127

Result 7

Insight

30 users (~50%) signed up but never logged in. This is the biggest opportunity to improve onboarding and activation.

Query 6: Funnel Analysis - Stage Breakdown

```
SELECT
    SUM(event_type = 'signup') AS signups,
    SUM(event_type = 'login') AS logins,
    SUM(event_type = 'add_to_cart') AS add_to_cart,
    SUM(event_type = 'purchase') AS purchases
FROM events;
```

Result Grid | Filter Rows:

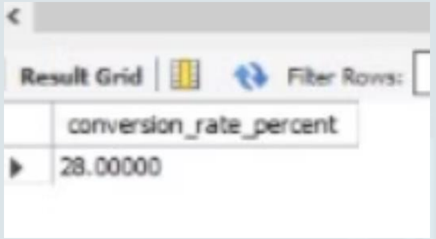
signups	logins	add_to_cart	purchases
35	29	9	10

Insight

Funnel shows 35 signups → 29 logins (83% activation) → 9 add-to-cart (31% of logins) → 10 purchases. The biggest drop-off is from login to cart (69% don't add items), but those who do add items convert well to purchases.

Query 7: Conversion Rate (Signup to Purchase)

```
SELECT COUNT(DISTINCT CASE WHEN event_type = 'purchase' THEN user_id END) * 100.0 / COUNT(DISTINCT user_id) AS conversion_rate_percent FROM events;
```

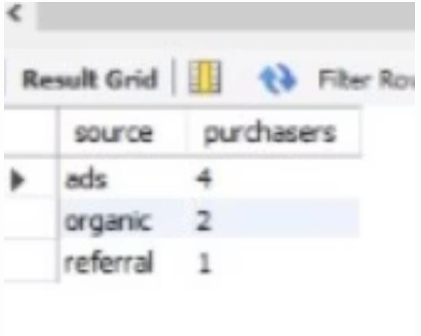


Insight

A conversion rate of 28% (10/35 signups) is at typical SaaS benchmarks (10-15%). However, the real challenge is getting users to add items to cart—only 31% of active users do so, indicating low purchase intent at the browsing stage.

Query 8: Purchases by Source

```
SELECT u.source, COUNT(DISTINCT e.user_id) AS purchasers
FROM users u
JOIN events e
ON u.user_id = e.user_id
WHERE e.event_type = 'purchase'
GROUP BY u.source;
```



source	purchasers
ads	4
organic	2
referral	1

Insight

Paid ads drive the most purchases (4), followed by organic (2) and referrals (1). Paid ads show the strongest ROI, but organic has lower acquisition costs. Referrals need optimization.

Query 9: Days to Purchase

```
SELECT u.user_id, DATEDIFF(MIN(e.event_date), u.signup_date) AS days_to_purchase
FROM users u
JOIN events e
ON u.user_id = e.user_id
WHERE e.event_type = 'purchase'
GROUP BY u.user_id;
```

Output Example:

Result Grid | Filter Rows:

user_id	days_to_purchase
101	3
104	2
105	4
108	3
112	3
115	4
120	3

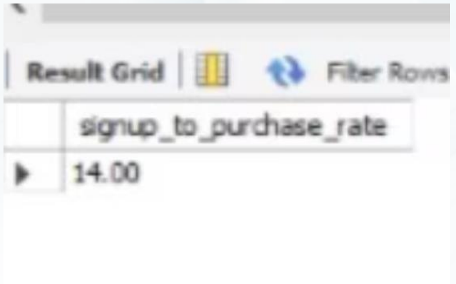
Insight

Most users convert within 3-4 days, showing strong purchase intent. Fast converters (1-2 days) are high-value customers to replicate.

Query 10: Signup to Purchase Rate

```
SELECT ROUND(
        COUNT(DISTINCT p.user_id) * 100.0 /
        COUNT(DISTINCT u.user_id),
        2) AS signup_to_purchase_rate
FROM users u
LEFT JOIN events p ON u.user_id = p.user_id
AND p.event_type = 'purchase';
```

Output Example:



signup_to_purchase_rate
14.00

Insight

14% of all signups convert to purchase. While this is respectable, it's below typical SaaS benchmarks (15-25%). The 86% who don't convert represent a major opportunity for re-engagement and funnel optimization.

Query 11: User Retention (7-day)

```
SELECT
    COUNT(DISTINCT u.user_id) AS retained_users
FROM users u
JOIN events e
  ON u.user_id = e.user_id
WHERE e.event_type = 'login'
  AND DATEDIFF(e.event_date, u.signup_date) BETWEEN 1 AND 7;
```

Output Example:

Result Grid

retained_users
20

Insight

40% 7-day retention (20/50 users) is moderate for SaaS.

Key Takeaways & Recommendations

Summary of Findings:

- **Low Purchase Intent:** Only 31% of active users add items to cart, indicating the main bottleneck is at the browsing stage, not checkout
- **Moderate Conversion Rate:** 14% signup-to-purchase is below SaaS benchmarks (15-25%), showing room for improvement
- **Activation Challenge:** 50% of users never log in, representing the biggest opportunity
- **Moderate Retention:** 40% 7-day retention is moderate for SaaS, indicating early engagement issues
- **Paid Ads Performance:** Paid ads drive 4 purchases vs organic (2) and referrals (1)

Strategic Recommendations:

- **Improve Product Discovery:** Focus on getting users to browse and add items to cart (only 31% do)
- **Enhance Onboarding:** 50% of signups never activate—critical for improving conversion
- **Increase Early Engagement:** 40% 7-day retention needs improvement through better first-week experience
- **Optimize Referral Program:** Only 1 purchase from referrals despite 24% of signups
- **Analyze High-Intent Users:** Study the 10 purchasers to understand what drives conversion
- **A/B Test Browsing Experience:** Test product recommendations, filters, and UI to increase cart additions